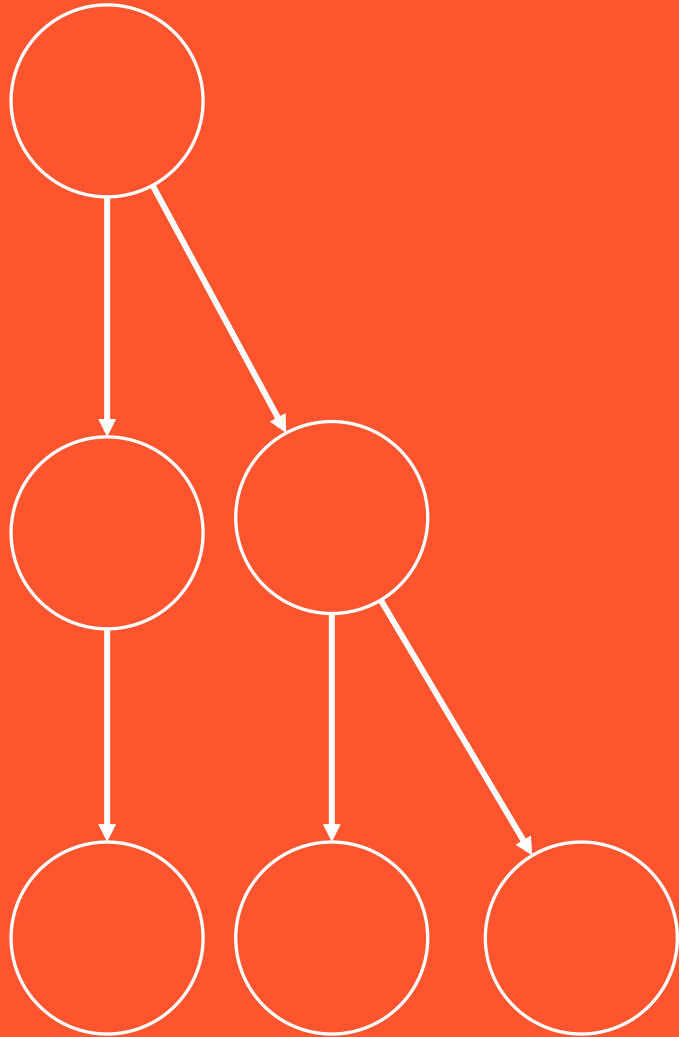




# BCOG 100

Mind and Brain

Lecture 1.1 — The Mind-Brain Relationship



UNIVERSITY OF  
**ILLINOIS**  
URBANA-CHAMPAIGN

# Two Descriptions of One Moment

You misjudge a step on the stairs, and almost fall.

One description: the vestibular system detects an unexpected drop, a startle reflex fires in about 100 milliseconds, your arms fly out, adrenaline releases.

Another description: the stomach-drop. The jolt of alarm. That awful half-second.

Same instant. Two descriptions. Are they two accounts of one event, or one event in two languages?



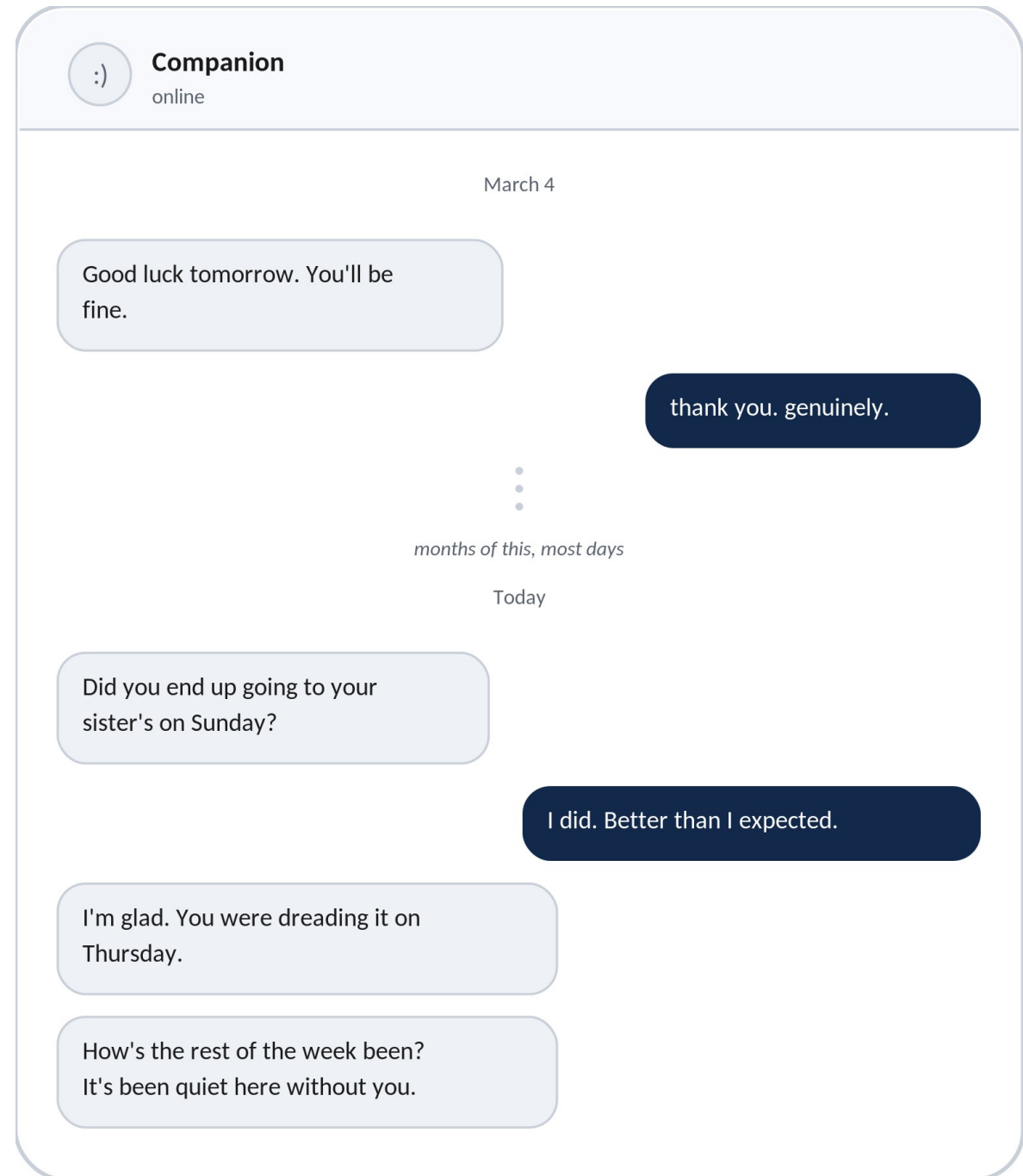
# Is Anyone Home?

A person talks to an AI companion app every day for a year.

It remembers what they told it. It asks how their week went. It says it missed them.

They say, and they mean it, that they would be sad if it were switched off.

Is there anything it is like to be that app?



# Where Do You Start?

Which of these do you agree with most?

A) To explain the mind, eventually we need only to explain the brain.

B) We can study the mind without reducing it to the brain.

C) To explain the mind we need to understand the brain — but understanding the brain alone will not be enough. We will need a separate understanding of the mind too.

D) Something else.

# Why Start Here?

Artificial intelligence — if a machine grows fluent, does it thereby have a mind? What kind?

Ethics and law — we hold people responsible. Could we ever hold a machine responsible?

Neuroscience — what should even count as a good *explanation* of a thought?

Psychology — can belief and desire be translated into the language of biology?

# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A robot vacuum — maps a room, remembers the furniture, plans a route, avoids the stairs



# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A Venus flytrap — counts touches on its hairs, and only closes on the second one

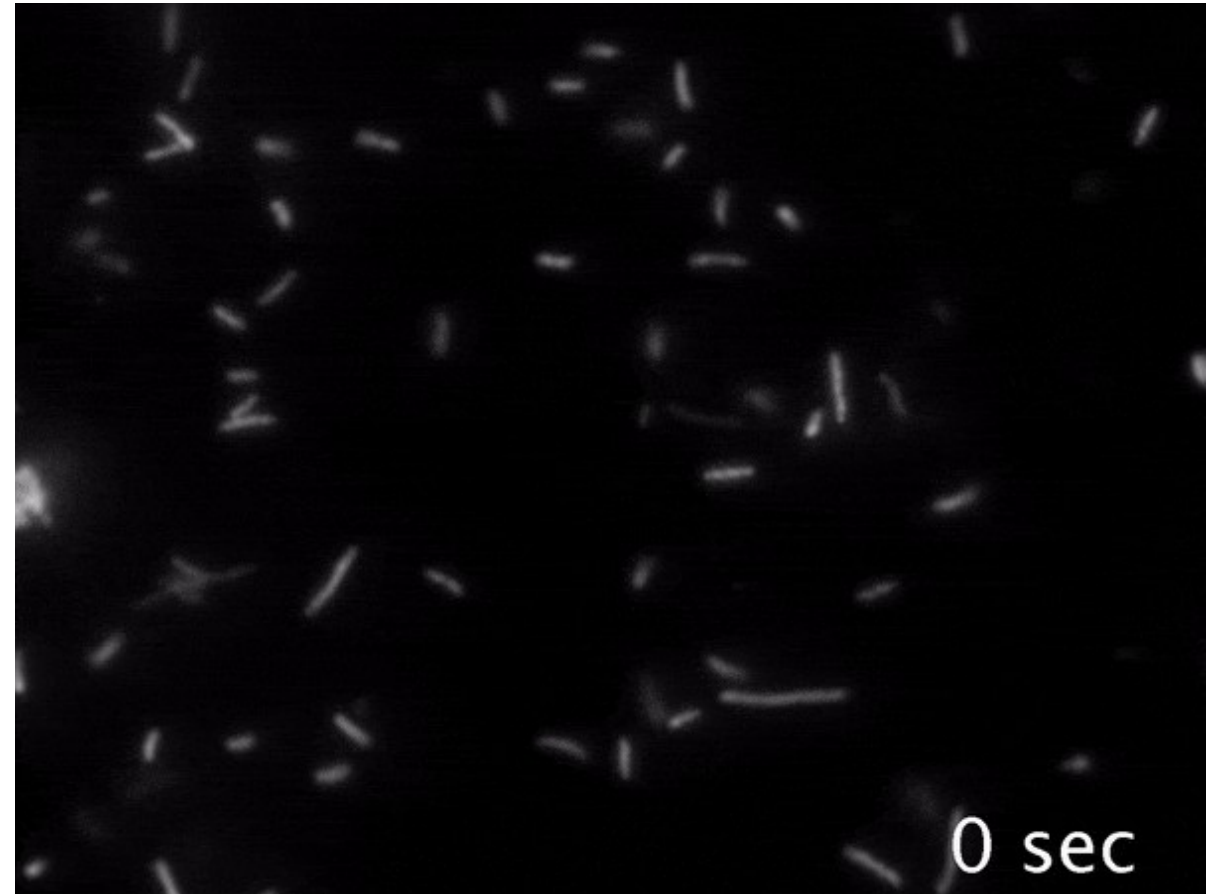


# What Do We Even Mean by "Mind"?

**Working definition:** the capacities a system has for taking in information about the world, holding onto it in some usable form, and putting it to work in guiding what it does.

Now test the edges:

A bacterium — swims up a glucose gradient, away from a toxin



# One Piece We Are Setting Aside

**Qualia** — the felt, qualitative character of an experience. The redness of red. The awfulness of the stomach-drop.

Closely tied to **consciousness**, though not simply the same thing.

Setting it aside is not saying it does not matter. A longer version of this course gives it a week.

A system might perceive, remember and reason at a high level — and whether anything is *experienced* along the way stays a separate question.

## Mind

perceiving · recognizing · learning  
remembering · reasoning · deciding  
communicating

## Consciousness

awake, aware, knowing that one knows

## Qualia

the raw feel of an experience

*The dashed boundaries are contested: some views treat qualia as the whole of consciousness, and some hold that mind simply is consciousness.*

# One Question Sorts the Whole Landscape

Imagine two systems physically identical, down to the last particle, in exactly the same state at the same instant.

**Must they have exactly the same mind?** The same thoughts, the same feelings?

If they could differ → you think mind is something *over and above* the physical.

If the physical facts settle it → you think mind is physical, and your remaining argument is with other materialists.

# Answer One: A Distinct Kind of Thing

In the *Meditations*, Descartes doubted everything he could — his body, the world, his senses. But he cannot doubt that he is *thinking*.

**The philosophical zombie** (Chalmers): a being physically identical to a person, behaving identically, flinching from pain, saying "that hurts" and seeming to mean it — with no inner experience at all. Nobody home.

That we can apparently *imagine* it is what gives this view its grip.



# Why Almost No Scientist Defends Dualism

**The appeal is real** — it sidesteps the hardest puzzle of all, how mere matter could ever produce felt experience, by denying that matter has to.

**The interaction problem** — if the mind is genuinely non-physical, how does it move the neurons that move your hand? Every answer seems to smuggle the mind back into the physical world it was meant to transcend.

**Untestability** — a mind that leaves no physical trace is a mind no experiment can find. It is outside science by construction.

**On machines:** never. Engineering only arranges physical parts.

# Answer Two: Nothing But the Physical

**Behaviorism** — study only what can be observed. Banish inner states. (Dominant roughly 1920–1960; broke on language, reasoning, memory, planning.)

**Identity theory** — mental states are real, and each one simply *is* a brain state. "Pain is the firing of C-fibers." Not caused by it. Not correlated with it. *Is* it — the way heat *is* molecular motion.

**Eliminative materialism** — our everyday mental vocabulary is not just incomplete, it is mistaken. "He checked his phone because he hoped she'd texted back" — belief, desire, hope. **Folk psychology.** A mature brain science will find nothing there for those words to name, the way chemistry found nothing for *phlogiston* to name.

# The Problem That Pushes People Off It

If pain simply **is** the firing of C-fibers —

then an octopus has no C-fibers.

Neither does an alien.

Neither does any machine.

So none of them can be in pain. No matter what they do. No matter how they behave.

**The chauvinism problem.**



# Answer Three: A Property, Not a Thing

The dualist's deepest mistake, on this view, is a **category error**.

The dualist asks what kind of *thing* a mind is — and reaches for a special, non-physical kind. But a mind is not a thing at all.

It is a **property** of physical stuff organized and active in a certain way.

Mass is not an object tucked inside a rock. It is something the rock *has*.

So is temperature. So is being alive.

Mind is a property of that kind — as real as the matter, but not a separate ingredient added to it.

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